

# Tristan Surajbali

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## Education

### University of Wisconsin-Madison

*Expected May 2028*

*Bachelor of Science, Electrical Engineering*

GPA: 3.55/4.0, Dean's List (3x), Mercile J. Lee Scholar, STAR Scholar

## Engineering Experience

### Madison Gas & Electric

*June-Aug. 2026*

#### *Renewable Engineering Intern*

- Streamlined solar maintenance workflows by developing remote-access procedures for tracker and inverter controls and implementing equipment tracking for 5,000+ renewable-energy assets, reducing maintenance time by 40%
- Led root-cause investigations across 10+ solar sites using SCADA monitoring, thermal imaging, drone inspection data, and field testing to identify equipment failures and inform corrective maintenance
- Commissioned EV charging equipment across a fleet of 140+ public chargers, executing functional test charges and verifying hardware and network connectivity for field deployment
- Analyzed BESS control systems through Tesla Megapack O&M and RTAC troubleshooting, supporting maintenance of utility-scale storage controls and monitoring systems

### Badger Solar Racing

#### *Low Voltage Project Lead*

- Lead technical development of the low voltage system, serving as primary point of contact for design decisions, debugging, and mentoring members throughout PCB development, testing, and system integration
- Revised the Power Distribution & Control (PDC) board schematic and PCB in Altium, integrating regulated power distribution, CAN communication, current/voltage telemetry, circuit protection, and motor-controller interfaces
- Optimized the 12V-to-5V power stage through component trade studies, selecting a 3 A synchronous buck converter that reduced circuit footprint ~60%, reclaiming board space for current-sensing telemetry and input voltage regulation
- Simulated and validated voltage-regulation circuitry in Spice, evaluating MOSFET alternatives and circuit behavior before PCB implementation, bring-up, and bench testing
- Resolved electrical faults across CAN and lighting systems using fault-tree analysis and DMM continuity/impedance measurements, resulting in 200+ competition miles with zero LV failures and team's first ASC qualification

## Projects

### *Temperature-Controlled Fan*

- Designed a custom STM32 PCB integrating protected 12 V input, buck regulation, temperature sensing, and MOSFET PWM control of a 6 W brushless fan
- Implemented temperature-based control and current fault detection, validating power, PWM, and sensing circuitry through PCB bring-up and oscilloscope testing

## Leadership Experience

### Engineering Student Scholars Program Leadership Board

#### *Emerging Leader*

- Represent engineering students as 1 of 15 selected leaders, coordinating community and academic initiatives and providing student feedback directly to College of Engineering leadership
- Mentor undergraduate engineering students on academics, professional development, and navigating UW engineering through individual guidance and ongoing peer support

### Redefined Acapella

#### *Music Director*

- Direct musical operations for a 20-member a cappella group, leading 6 hours of weekly rehearsals and preparing 20+ songs for 5+ gigs and a semester concert
- Teach and arrange repertoire, instructing 20+ songs while providing feedback to improve ensemble performance
- Lead artistic selection and auditions, evaluating prospective members and coordinating group decisions on new members, repertoire, and semester performance priorities

## Skills & Interests

**Programming:** C, Python, Java, Verilog

**Software:** Spice, Altium Designer, MATLAB, Microsoft Office 365

**Hardware:** PCB Design & Assembly, Soldering, Oscilloscopes, Multimeter testing, Debugging, CAN, I2C, isoSPI, USB

**Relevant coursework:** Circuit Analysis, Signals & Systems, Digital Systems Fundamentals

**Interests:** Volleyball, Rock climbing, Soccer, Guitar, Cubing